I changed experiment 4 BIST so that it tests the even addresses first and then the odd addresses. In each session, it writes in increasing order, then reads back in decreasing order and checks every value. A new flag, even, tells the state machine which session it is in. The value written is the low 16 bits of the address XORed with A5A5 for even addresses and 5A5A for odd addresses.

The SRAM has a 2-cycle latency, so the data on the bus at any moment belongs to the address I sent two cycles earlier. Since the address changes by 2 each step, the expected value in S_READ_CYCLE is the value for BIST_address + 4. The states S_DELAY_1 and S_DELAY_2 start the reads after the last write. After the last read is issued, S_DELAY_3 and S_DELAY_4 check the final two returned values (addresses 2 and 0, or 3 and 1). Each uses a different offset in the even and odd sessions, because the last read address is 0 in the even session and 1 in the odd session. BIST_finish is set only in S_DELAY_4 of the odd session, after the read of address 1 has been checked.

In the even session, S_DELAY_3 and S_DELAY_4 do not write anything. At the end of S_DELAY_4 I set BIST_address to 1 and start the odd session's writes there. This way each location is written once and read once, and odd addresses are only written in the odd session.

The compare I kept in S_WRITE_CYCLE only works for the first writes because it assumes the SRAM starts at all zeros, so addresses below 4 are expected to read 0.

In simulation, BIST_mismatch stayed at 0, and the dumped SRAM contents matched the specified pattern at all locations.